

HEALTHCONNECT CLINIC

Advanced Analytics & Decision Support Report

Data Analytics Track

1. Week 5 → Week 6 Transition

Week 5 established the HealthConnect appointment dashboard, core attendance/no-show KPIs and exploratory analysis. Week 6 builds forward by validating key findings, investigating relationships between risk factors, and translating results into decision support rather than repeating the Week 5 analysis.

The main Week 6 focus is booking lead time, previous no-show history and reminder channels, with appointment volume considered alongside rates so small groups are not over-interpreted.

2. Objectives

- Validate important Week 5 findings.
- Investigate deeper relationships and patterns associated with appointment no-shows.
- Compare important segments using rates and appointment volumes.
- Identify high-impact findings relevant to clinic operations.
- Translate findings into practical recommendations.
- Improve the existing Power BI dashboard with advanced analysis and decision support.

3. Advanced Analysis

3.1 Booking Lead Time Analysis

Booking lead time was grouped into 0–7 Days, 8–14 Days, 15–30 Days and 31+ Days. The observed no-show rate increases substantially across these groups.

Booking Lead Group	No-Show Rate	Total Appointments
0–7 Days	27.81%	640
8–14 Days	33.55%	602
15–30 Days	43.21%	1,333
31+ Days	60.49%	2,425
Total	48.46%	5,000

3.2 Previous No-Show History

Previous no-shows were grouped into 0, 1, 2 and 3+. The observed no-show rate increases with previous no-show history.

Previous No-Show Group	No-Show Rate	Total Appointments
0	43.51%	2,921

1	53.49%	1,548
2	59.36%	438
3+	68.82%	93
Total	48.46%	5,000

3.3 Previous No-Shows × Booking Lead

A matrix was created to investigate the combined relationship between previous no-show history and booking lead time. For patients with no previous no-shows, the observed no-show rate rises from 21.77% at 0–7 days to 55.18% at 31+ days. The broader matrix supports further investigation of this interaction, while smaller cells require caution.

3.4 Previous No-Shows × Reminder Channel

A second matrix examines reminder channel patterns across previous no-show groups. The overall attendance chart shows SMS at about 49.6%, Email 46.5%, WhatsApp 44.6% and None 42.7%. These differences indicate an area for further testing; they should not be treated as proof that the reminder channel itself causes improved attendance.

4. KPI & Finding Validation

KPI / Finding	Week 6 Validation	Decision
No-Show Rate	Reviewed against segmented and combined analyses.	Retain
Attendance Rate	Reviewed alongside no-show and reminder patterns.	Retain
Booking Lead Time	Strong increase in observed no-show rate with longer lead time.	Retain
Previous No-Show History	Higher previous no-show history corresponds with higher observed no-show rates.	Retain
Reminder Channel	Differences observed; causal effectiveness requires further testing.	Qualify

5. Validated Findings

Finding 1 — Booking Lead Time

Appointments booked 31+ days in advance recorded a 60.49% observed no-show rate across 2,425 appointments, making this a major operational risk segment.

Finding 2 — Previous No-Show History

Observed no-show rate increased from 43.51% among patients with no previous no-shows to 68.82% among the 3+ group.

Finding 3 — Combined Risk Segmentation

The combined analysis indicates that booking lead time and previous no-show history can be considered together when identifying higher-risk appointment segments.

Finding 4 — Reminder Channels

SMS showed the highest observed attendance rate among the displayed channels, but further testing is required before making a causal or permanent channel decision.

Finding 5 — Rate and Volume

Appointment volume should be considered alongside rates because very small groups can produce unstable percentages.

Key Findings

The Week 6 analysis strengthened the understanding that appointment no-show behavior is associated with more than one factor. Booking lead time and previous no-show history emerged as important analytical dimensions for identifying higher-risk segments.

- No-show rates increased across the booking lead groups, with appointments booked further in advance showing higher observed no-show rates.
- Patients with a history of previous no-shows generally showed higher current no-show rates than patients with no previous no-shows.
- The combined Booking Lead Group × Previous No-Show Group analysis provided a more useful risk view than examining either variable independently.
- Reminder channel analysis showed differences in attendance outcomes across SMS, Email, WhatsApp, and appointments without a reminder.
- Some smaller segments showed extreme rates; these should be interpreted alongside appointment volume rather than treated as strong conclusions on rate alone.
- The findings support further investigation into whether booking lead time and previous no-show history can help identify appointments that may require additional intervention.

6. Business Impact Assessment

Finding	Potential Business Impact	Priority
31+ day booking lead	Potential inefficient use of appointment slots; stronger confirmation may be warranted.	High
Previous no-show history	Supports targeted follow-up and risk segmentation.	High
Reminder-channel differences	Opportunity to investigate and optimise reminder strategy.	Medium
Small high-risk subgroups	Useful for monitoring but requires cautious interpretation.	Medium

7. Evidence-Based Recommendations

- Prioritise appointments booked 31+ days in advance for stronger confirmation and reminder procedures.
- Use previous no-show history as a risk indicator when considering which appointments may need additional follow-up.
- Further test reminder-channel performance, especially the observed SMS attendance pattern, before making a permanent decision.
- Use booking lead time and previous no-show history together as candidate risk-segmentation variables.
- Monitor no-show outcomes before and after interventions so operational changes can be evaluated.

9. Cross-Track Data Science Integration

Collaboration Attempt

As part of the Week 6 cross-track collaboration requirement, the Data Analytics track was expected to engage with the Data Science track to exchange findings and explore how the analytical results could support predictive modelling.

An attempt was made to identify and collaborate with a Data Science intern. However, I was unable to secure an available Data Science intern for a completed collaboration within the Week 6 timeframe. As a result, no formal cross-track integration or model level implementation could be completed during this week.

Proposed Data Science Contribution

Although direct collaboration could not be completed, the Week 6 analysis identified several findings that could provide a useful starting point for future Data Science work.

The main variables and relationships recommended for further investigation are:

- **Booking Lead Days** - longer booking lead times showed higher observed no-show rates.
- **Previous No-Shows** - patients with previous no-show history showed higher observed no-show rates.
- **Reminder Channel** - differences were observed across reminder channels and could be investigated as a potential predictive feature.
- **Booking Lead × Previous No-Show History** - the combined analysis provides a potential interaction to investigate in a predictive model.

These variables could be assessed by the Data Science track through feature selection, feature engineering and predictive modelling.

Potential Data Science Use Case

The key proposed use case is the development of a no-show risk prediction model that could estimate the likelihood of an appointment resulting in a no-show.

The Week 6 analytical findings could inform the selection of candidate features for such a model. In particular, the interaction between booking lead time and previous no-show history could be tested to determine whether combining these variables provides additional predictive value compared with using each variable independently.

The proposed modelling process could include:

- Selecting relevant appointment and patient-history variables.
- Testing booking lead time and previous no-show history as predictive features.
- Testing the interaction between booking lead time and previous no-show history.
- Evaluating the contribution of reminder channel.
- Comparing model predictions against actual appointment outcomes.
- Evaluating whether the model can effectively identify higher-risk appointments.
- Using model results to refine the operational recommendations developed by the Data Analytics track.

The absence of a Data Science collaborator meant that these proposed modelling activities could not be validated or implemented as part of Week 6 Data Analytics deliverable. Therefore, I'm presenting the recommendations in this section as proposed next steps rather than completed cross-track outcomes.

10. Assumptions, Limitations, Risks & Dependencies

- The analysis identifies associations rather than proving causation.
- Some combinations, especially within the 3+ previous no-show group, have relatively small volumes.
- Reminder-channel differences may be influenced by other patient or appointment characteristics.
- The dataset is historical and may not represent future behaviour.
- Predictive modelling and testing are needed to determine whether identified factors improve prediction or interventions reduce no-shows.
- Operational decisions should consider both rate and appointment volume.